User Management CRUD API

Backend Development Project 2 – Database Integration

Submitted By:

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Technologies Used:

Node.js | Express.js | MongoDB | Mongoose

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Introduction

The User Management CRUD API is a RESTful backend application developed using Node.js, Express.js, and MongoDB.

The application demonstrates database integration by performing Create, Read, Update, and Delete (CRUD) operations on user records while ensuring data persistence and preventing duplicate entries.

Objectives

The objectives of this project are:

- Connect an API to a database
- Design a User schema
- Implement CRUD functionality
- Store data permanently in MongoDB
- Prevent duplicate user entries
- Practice NoSQL database operations using Mongoose

Technologies Used

Technology	Purpose
Node.js	Runtime Environment
Express.js	Backend Framework
MongoDB	NoSQL Database
Mongoose	ODM for MongoDB
Postman	API Testing
VS Code	Development Environment

Project Structure

Task-2-Nosheen-Afzal/

```
│
├── models/
│   └── User.js
│
├── routes/
```

```
│   └── userRoutes.js
│
├── screenshots/
│
├── .gitignore
├── package.json
├── package-lock.json
├── server.js
├── README.md
└── Backend Task 2.pdf
```

Database Design

User Schema

```
{
  name: String,
  email: String,
  age: Number
}
```

Field Description

Field	Type	Description
name	String	User Name
email	String	Unique Email
age	Number	User Age

Validation

- Name is required
- Email is required
- Email must be unique
- Age is required

API Endpoints

Create User

POST

/api/users

Sample Request:

```
{
  "name": "Nosheen",
  "email": "nosheen@gmail.com",
  "age": 22
}
```

Get All Users

GET

/api/users

Get User By ID

GET

/api/users/:id

Update User

PUT

/api/users/:id

Delete User

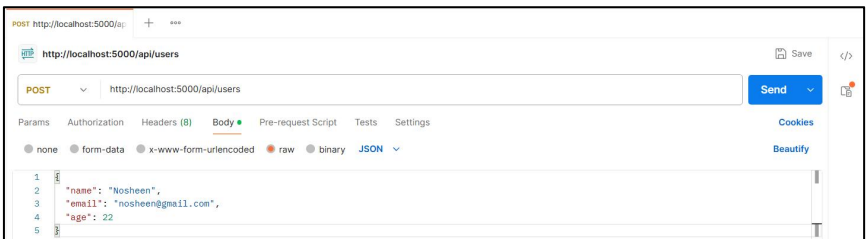
DELETE

/api/users/:id

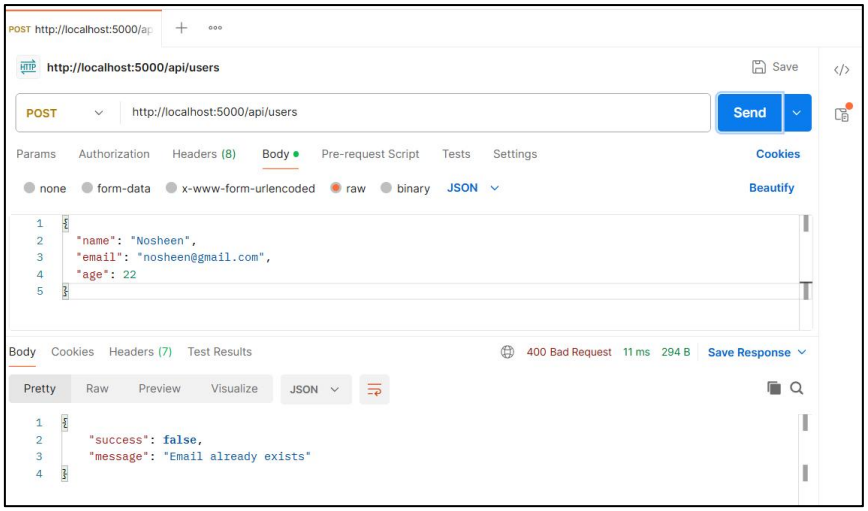
Testing & Results

MongoDB Connection

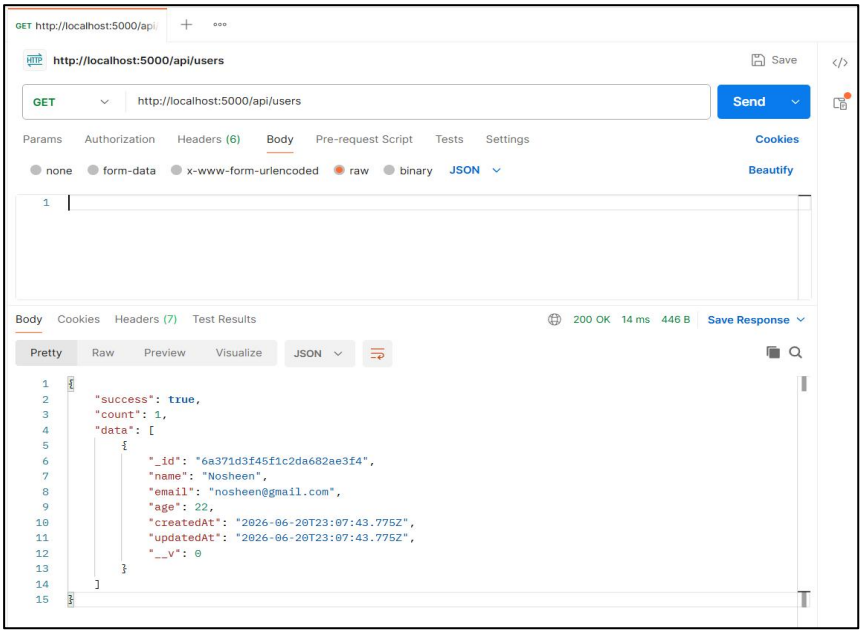
Create User Test



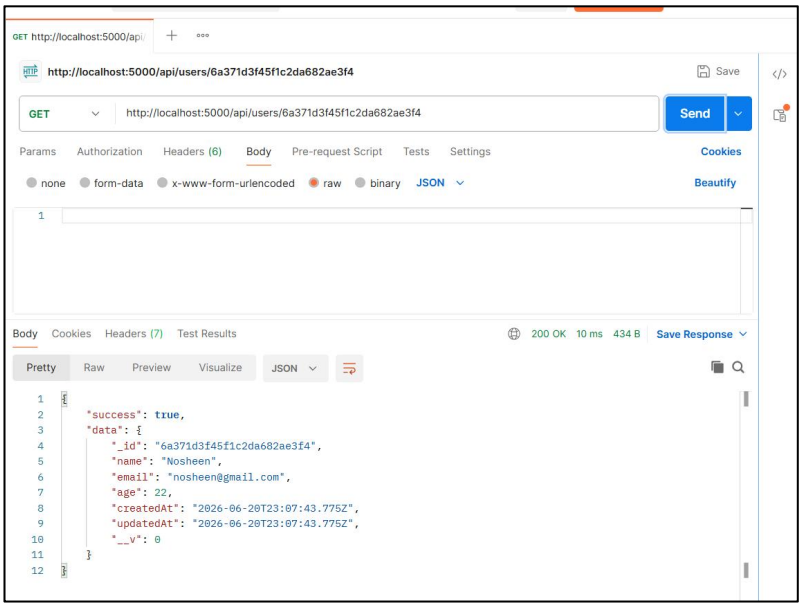
Duplicate Email Validation



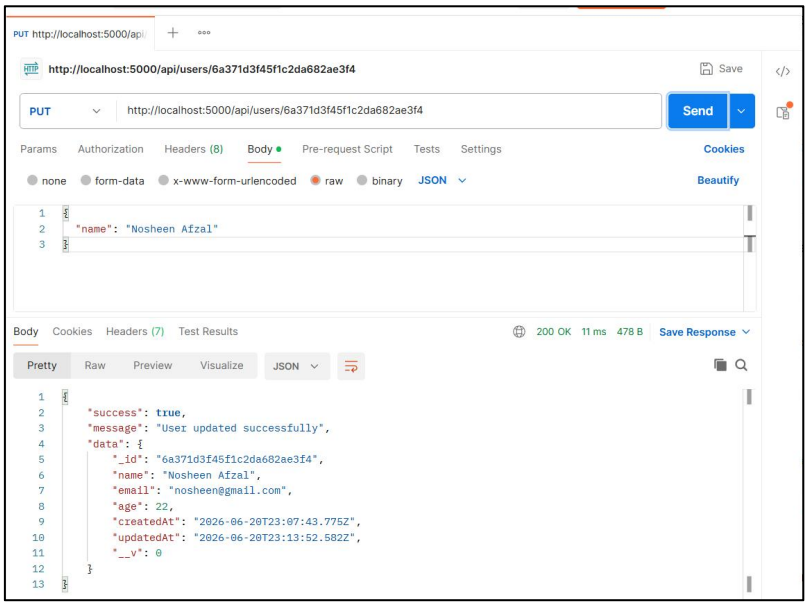
Get All Users



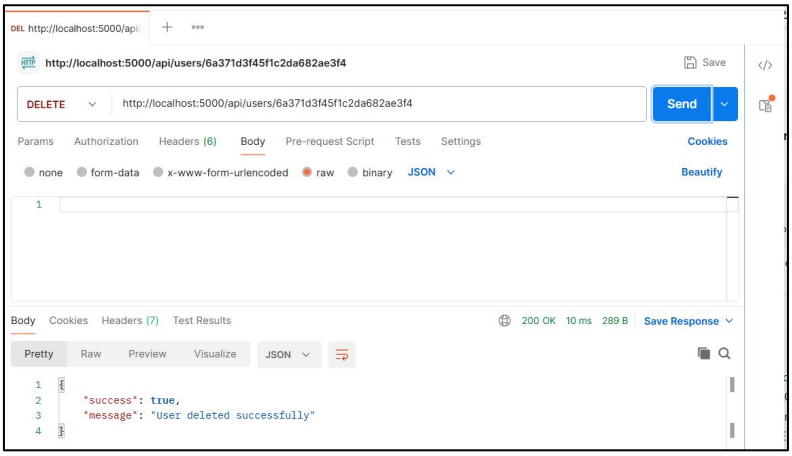
Get User By ID



Update User



Delete User



Conclusion

This project successfully implemented a RESTful API integrated with MongoDB using Mongoose. CRUD operations were developed and tested using Postman. The application demonstrates database connectivity, data persistence, schema design, and duplicate-entry prevention.